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Research article



Assessment of greenhouse gases emissions and intensity from Chinese marine aquaculture in the past three decades

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ABSTRACT

Marine aquaculture is increasingly gaining importance as a source of food with high nutritional value. However, the expansion of aquaculture could be responsible for water contamination that influences the environmental quality of coastal ecosystems, and emissions of greenhouse gases (GHG) that affect global climate. China is the world's largest producer of marine aquaculture protein, which demands robust studies to assess the corresponding GHG emissions and intensity. To fill in this knowledge gap, the current study quantifies and analyzes GHG emissions and intensity (emission intensity is defined as GHG emissions per unit of production) from Chinese marine aquaculture (marine aquaculture production) over the past 30 years (1991–2020). The production of marine aquaculture comes from the China Fisheries Statistical Yearbooks. And the GHG emissions and intensity were calculated based on five sectors (commercial feed, trash fish, N_2O, CH_4, and energy) by Emission-Factor Approach. The results suggest that, excluding shellfish and algae, GHG emissions of ten coastal provinces (excluding Shanghai, Hong Kong, Taiwan, and Macau) increased from 2 Mt (10^9 kg) CO_2-eq in 1991 to 25 Mt CO_2-eq in 2020. In contrast, GHG emission intensity decreased in the same period from 7.33 (t CO_2-eq/t production) to 6.34 (t CO_2-eq/t production), indicating a progressive mitigation in GHG emissions per unit of product, hence sustainably satisfying a growing demand for food. As a result, China's marine aquaculture seems to be paving a promising way towards the neutrality of GHG emissions. In most provinces, GHG is on the rise, and only in Tianjin is on the decline in recent years. For the emissions intensity, the values of more than half provinces showed the downtrends. In addition, by considering the ratio of shellfish and algae, Chinese marine aquaculture can improve the net zero goal for GHG emissions of the sector. Finally, results also reveal for the first time the changes in taxonomic composition and spatial GHG emissions and intensity, providing new understanding and scientific bases to elaborate consistent mitigation strategies for an expanding global marine aquaculture.

1. Introduction

Global aquaculture has grown rapidly over the past decades because aquatic products have gained relevance as a source of food with high nutritional value (Naylor et al., 2021). Global aquaculture production increased from 34.2 Mt (10^9 kg) in 1997 to 82.1 Mt in 2018 (FAO, 2019; FAO, 2020) and will remain growing significantly. The number of cultured species is also increasing, with more than 300 species in 2017,

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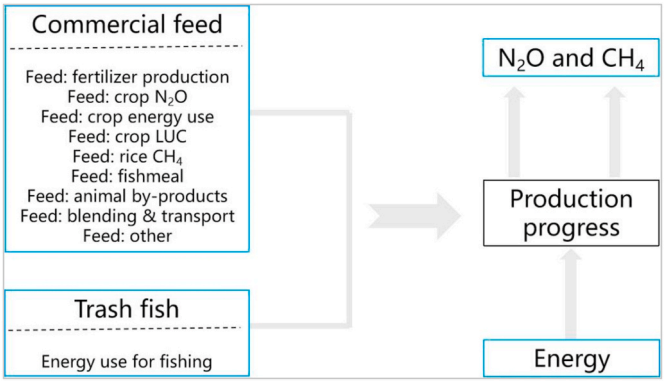


Fig. 1. Scope graph of marine aquaculture production contributing to greenhouse gas emissions in this study (the blue frames represent the sectors that were calculated).

including fish, crustaceans, aquatic plants, and mollusks (Cao et al., 2013; Tacon, 2019). However, the expansion of aquaculture potentially degrades the quality of the environment and the climate (Gephart et al., 2021) through e.g. land-use changes (LUC), biodiversity loss, and emissions of greenhouse gases (GHG) (Diana, 2009; Chang et al., 2017; Arifanti et al., 2019).

An increase in GHG emissions can cause a series of environmental problems (Doney et al., 2012). For instance, GHG generated from human activities has been identified as the most significant driver of observed climate change since the mid-20th century (IPCC, 2013). GHG emissions from aquaculture gather nitrous oxide (N_2O), carbon dioxide (CO_2), and methane (CH_4) (MacLeod et al., 2019). N_2O is a by-product of microbial nitrification and denitrification, with a global warming potential 310 times that of CO_2 (Hu et al., 2012; MacLeod et al., 2019). Many studies have shown that GHG emissions derive from all steps of aquaculture processes, but feed and energy use are both outstanding (Cao et al., 2011; MacLeod et al., 2019; Song et al., 2019). Especially for feed, the feed conversion ratio is the strongest level (Gephart et al., 2021). Besides, species is also an important factor in GHG emissions. By comparing the GHG emissions of different species, Gephart et al. (2021) find that farmed bivalves and seaweeds generate the lowest GHG emissions, in addition, silver and bighead carps have the lowest greenhouse gas among farmed finfish and crustaceans. Farming models can also affect the amount of GHG emissions. Liu et al. (2016) showed that the GHG emissions of land-based closed containment water recirculating aquaculture systems using a typical US electricity mix based on fossil fuels is twice that in traditional open net pen systems for salmon produced when delivery to the market is not included. Given the high volume of the world's aquaculture and its importance for high-quality protein supply, quantifying and mitigating GHG emissions from aquaculture is critical to the global effort for GHG emissions reduction.

Asia plays a big role in global aquaculture markets (Jiang et al., 2022). Led by China, Asia countries produced over 91% of global aquaculture deliverables in 2017 (Bostock et al., 2010; Tacon, 2019). As the largest aquaculture producer, with a long history and tradition (Li et al., 2011), China currently accounts for more than half of the global aquaculture volume (Garlock et al., 2020). Since 1978, aquaculture industry has developed rapidly during China's Reform and Opening Up (Wang et al., 2020). The total production of freshwater and marine aquaculture ranges from 6.53 Mt in 1991 to 52.24 Mt in 2020 (China Fishery Statistical Yearbook, 1991), increased 8-fold in the past 30 years. Concomitantly, the share of freshwater and marine aquaculture has been changing, with the share of marine aquaculture production gradually increasing from 29.10% in 1991 to 40.87% in 2020 (China Fisheries Statistical Yearbooks). The cultured species in China's marine aquaculture can be subdivided into six species-group: fish, shrimp, crabs, shellfish, algae, and other species, among which shellfish and algae are considered as carbon sinks (Tang et al., 2011).

Although the aquaculture industry is relatively young and currently produces only a small proportion of GHG compared with agribusiness industries, considering a scenario of rapid development, aquaculture may enhance its GHG emissions (Yuan et al., 2019; MacLeod et al., 2020; Tacon, 2019). Therefore, a comprehensive understanding of GHG emissions from aquaculture is increasingly important in the context of global warming and sustainable food production.

The emission intensity (EI) of GHG has been widely used as an index related to the amount of CO_2-equivalent (CO_2-e) emitted per unit, EI indicates the level of GHG emissions from a production system in which larger EI values correspond to higher CO_2-e emitted per unit. The units of EI can be defined according to the purpose. The unit of $kgCO_2$-e/kg production or $kgCO_2$-e/kg edible production was used most commonly (Iribarren et al., 2010; Aragão et al., 2022). For instance, Chang et al. (2017) used life cycle assessment to analyze the carbon footprint of an ecological shrimp farm by using a function unit per kilogram, and MacLeod et al. (2020) quantified GHG emissions intensity from global aquaculture by using the $kgCO_2$-e/kg edible production. Besides, kg CO_2-e/100 kcal, kg CO_2-e/kg protein, and t CO_2-e/hm^2 are also common units (Zhu et al., 2016; Ray et al., 2019; Kosten et al., 2020). Nonetheless, EI (CO_2-e/live weight production) of China's marine aquaculture is still unclear. To fill in this knowledge gap, we searched for useful datasets to quantify and analyze both GHG emissions and EI from China's marine aquaculture over the past 30 years (1991–2020).

In specific, the objectives of this study are to (1) quantify the changes in GHG emissions and EI of marine aquaculture over the past 30 years in China, (2) compare these changes from different provinces, and (3) analyze the contribution of shellfish and algae to the net GHG exchanges.

2. Materials and methods

2.1. Datasets

GHG emissions from China's marine aquaculture were estimated over the past 30 years by considering sectors as follows (Fig. 1):

(1) production of feed raw materials to processing and transport of feed materials to compound feed in feed mills, transportation to farms (CO_2, CH_4 and N_2O);
(2) trash fish (energy use);
(3) electric power use in ponds and farms (CO_2);
(4) residual bait, non-edible part of fish, shrimps, crabs and others (N_2O) (Macleod et al., 2020);
(5) residual bait and excrements (CH_4).

There are 14 provinces with marine aquaculture in the east, southeast, and south of China. These provinces are around the Bohai Sea, Yellow Sea and South China Sea, in East and South China, respectively, which have great potential for marine aquaculture development. To analyze the changes in GHG emissions, the species were identified by extracting the data for ten coastal provinces (Tianjin, Hebei, Liaoning, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, Guangxi, Hainan) over the past 30 years from the China Fisheries Statistical Yearbooks. Data for Shanghai, Hong Kong, Taiwan, and Macau were excluded because of missing values.

2.2. GHG emissions from feed

Inter-annual GHG emissions depend on four factors: non-feed rate (non-*FR*), feed conversion ratios (*FCRs*), feed ratio composition (*FRC*), and emission factors (*EFs*) for feed raw materials and non-commercial feed (trash fish). *FCR* varies as its application develops in response to commercial demand (MacLeod et al., 2019; Soto, 2019; Naylor et al., 2021). However, values are kept fixed for each species over 30 years due

to parameter limitations.

For the non-*FR*, *FCR*, and *FRC*, the data of inter-annual change are based on Tang et al. (2016), Greenpeace (2017), and MacLeod et al. (2020). According to Eq. (1), *FR* (%) is multiplied by the production (P_i, tonnes) for the corresponding species (i) and FCR_i (tonnes commercial feed/tonnes production)to derive the total weight of the compound feeds used (F_i).

$$F_i = FR_i * P_i * FCR_i \tag{1}$$

For the commercial feed, which was composed of different raw materials, the composition differs from regions, species, and years. Therefore, GHG emissions (GE_{fi}) of every raw feed material (i) were calculated by the weight of each raw feed material (RF_{fi}, tonnes) multiplied by the emission factor (EF_{fi}, tonnes CO_2-e/tonnes raw feed material) of each raw material (Eq. (2)) (Macleod et al., 2020). All GHG emissions of compound feed are then calculated through the FISH-e tool.[1]

$$GE_{fi} = EF_{fi} * RF_{fi} \tag{2}$$

The non-commercial feed mainly includes trash fish, mainly from non-domesticated species. In this study, the EF_{nf_y} (ton fuel/ton production) of non-commercial feed was calculated according to marine wild fisheries and diesel use (Xu et al., 2009) in which

$$EF_{nfy} = KW_y * b / P_y \tag{3}$$

where KW_y is the power used for the ship of each year y, b is the fuel consumption factor (tonne fuel/KW_y), and P_y is the fishing production. Because non-commercial feed mainly comes from domestic fishing (Greenpeace, 2017), power and production data are from domestic fishing. When there were no domestic fishing data, domestic fishing and pelagic fishing data were used. GHG emissions of non-commercial feed are shown in Eq. (4)

$$GE_{nfi} = FR_i' * P_i * FCR_i' * EF_{nfy} * EF_f \tag{4}$$

where GE_{nfi} is the value of GHG emissions from non-commercial feed for species i, FRi' and $FCRi'$ are feed rate and feed conversion ratio by non-commercial feed, EF_f (3.21 kg CO_2/kg fuel) is the emission factor of fuel (Yue et al., 2013).

2.3. GHG emissions from power supply

Electricity is used on farms primarily for pumping water, aerator, and feeding machine. The amount of energy can be calculated based on pond areas and farm areas (Xu et al., 2007). For the pond, the amount of electrical energy (*KWh*) for the aerator was determined by the areas of the aerator (*AR*), the power of each acre (*PA*, 0.15kw/acre), the time used per day (T, 4 h) and the number of days (d, 200 days), and the annual (y) consumption energy was calculated as (Xu et al., 2007)

$$KW \cdot h_y = AR_y * PA_y * T_d * d_y \tag{5}$$

Annual GHG emissions of the aerator (GE_{peay}) were calculated as (Liu and Che, 2010)

$$GE_{peay} = KW \cdot h_y * C * E * CC * CF * TP_y \tag{6}$$

where C (0.356) is the weight of coal converted into electricity per *kWh*, E (0.982) is the effective oxidation fraction of coal, *CC* (0.73257) is carbon content in coal, *CF* (3.67) is a conversion factor, and TP_y is the annual thermal power generation ratio.

For the pumping water, GHG emissions were calculated as

$$GE_{pepy} = \left(V_y * WER * d_y + V_y\right) / b_1 * C * E * CC * CF * TP_y \tag{7}$$

where V_y is the volume of aquaculture water, *WER* is the daily water change ratio (2%), b_1 is 60 cubic meter/kw, which means the amount of water output per *kW* of electricity.

For the aerator in factory farms (Xu et al., 2007), the GHG emissions were determined as

$$GE_{feay} = V_y * d_y * de_y * dfr * OC/OP * C * E * CC * CF * TP_y \tag{8}$$

where de_y is the average breeding density of every year, *dfr* is the daily feeding rate (3%), *OC* is oxygen consumption (0.5 kg oxygen/kg feed), *OP* is oxygen production (0.8 kg oxygen/kwh).

For the pumping water in factory farms (Xu et al., 2007), GHG emissions were determined by the volume of aquaculture water V_y, the number of water turnover cycles per day (n, 4), d_y (200 days) and b_2 (20 cubic meter/kw) (Eq. (9)).

$$GE_{fepy} = V_y * n * d_y / b_2 * C * E * CC * CF * TP_y \tag{9}$$

For the feeding machine, GHG emissions were calculated by the power consumption of the feeding machine based on the operating time of the feeding machine (Eq. (10)).

$$GE_b = Ar/0.8 * KW * T * C * E * CC * CF * TP_y \tag{10}$$

In Eq. (10), *Ar* is the breeding area, *KW* is the power for each feeding machine (0.075kw), and 0.8 is the feeding area of each feeding machine (Wu, 2021).

2.4. Annual N_2O emissions

N_2O emissions are determined by many factors like temperature, salinity and pH, and vary with instantaneous environmental conditions (Hu et al., 2012). For that reason, it is not an easy task to quantify N_2O emissions from aquaculture. Despite these difficulties, MacLeod et al. (2019) calculated the N_2O *EF* (0.791 kgCO_2e/kg live weight, LW, production), where CO_2-e represents the amount of other GHGs relative to CO_2 GWP, so aquatic integrated N_2O emissions are calculated by multiplying the production by N_2O *EF* per kg of production (Eq. (11)).

$$GE_{Ni} = EF_{Ni} * P_i \tag{11}$$

Where GE_{Ni} is N_2O GHG emissions in CO_2-e, EF_{Ni} is the 0.791 kgCO_2-e/kg LW production (MacLeod et al., 2019), and P_i is the production per species group.

2.5. Annual CH_4 emissions

CH_4 has no negligible impact on aquaculture GHG emissions. The main sources are the excess feed and animal excrement. CH_4 emission is a relatively complex process but largely affected by dissolved oxygen content in water. For this study, based on Ma et al. (2018) and Wu (2021), CH_4 emissions are accounted for as (Eq. (12)).

$$GE_{CH4} = A_r * P_c \tag{12}$$

Ar is the aquaculture area, P_c is the CH_4 emission coefficient per unit aquaculture area (51.6 kg/hm^2), where hm^2 corresponds to a hectare or 10,000 m^2.

2.6. GHG emissions from China's marine aquaculture

The integrated annual GHG emissions (GE_y) are defined as the sum of GHG emissions for feed (GE_f and GE_{nf}), GHG emissions for energy use (GE_{peay}, GE_{pepy}, GE_{feay}, and GE_{fepy}), GHG emissions for aquatic N_2O (GE_N) and GHG emissions for CH_4 (GE_{CH4}) (Eq. (13)). All of the amounts of GHG were standardized relative to CO_2-e (CO_2 global warming potential, GWP).

[1] http://www.fao.org/fishery/affris/affris-home/fish-e-faos-tool-for-quantifying-the-greenhouse-gas-emissions-arising-from-aquaculture/en/.

Table 1
The species of shellfish that were cultured and their parameters source.

Species	R_{ST}	W_C	R_S	W_{CS}	Parameters source
Cupped oysters nei (*Crassostrea* spp)	1.30%	44.90%	63.80%	11.50%	Tang et al. (2011)
Abalones nei (*Haliotis* spp)	4.43%	44.99%	56.45%	11.98%	Gu and Zhu (2021) (mean value of other shellfish)
Chinese mystery snail (*Cipangopaludina Chinensis*)	4.43%	44.99%	56.45%	11.98%	Gu and Zhu (2021) (mean value of other shellfish)
Blood cockle (*Anadara granosa*)	3.66%	45.86%	34.33%	11.29%	Zhou et al. (2002); Tang et al. (2011)
Sea mussels nei (*Mytilidae*)	4.63%	46%	70.64%	12.70%	Tang et al. (2011)
Pen shells nei (*Atrina* spp)	2.30% (percentage of live weight)		7.02% (percentage of live weight)		Guo (2015)
Scallops nei (*Pectinidae*)	7.32%	43.90%	56.58%	11.40%	Tang et al. (2011)
Japanese carpet shell (*Ruditapes philippinarum*)	7.67%	42.80%	44.65%	11.40%	Tang et al. (2011)
Constricted tagelus (*Sinonovacula constricta*)	6.62%	44.99%	64.78%	13.24%	Lv et al. (2014); Ke et al. (2016)

Table 2
The species of algae that were cultured and their parameters source.

Species	W_{CW}	Parameters source
Japanese kelp (*Laminaria japonica*)	31.20%	Lapointe et al. (1992); Zhou et al. (2002)
Wakame (*Undaria pinnatifida*)	30.70%	Guo (2015); Zhang et al. (2022)
Nori nei (*Porphyra* spp)	27.39%	Lapointe et al. (1992); Gao and Mckinley (1994)
Gracilaria seaweeds (*Gracilaria verrucosa*)	20.60%	Zhou et al. (2002); Zhang et al. (2022)
Fusiform sargassum (*Sargassum fusiforme*)	27.76%	Ji and Wang (2015); Zhang et al. (2022)
Eucheuma seaweeds nei (*Eucheuma* spp)	27.76%	Ji and Wang (2015); Zhang et al. (2022)
Bright green nori (*Enteromorpha clathrata*)	27.76%	Ji and Wang (2015); Zhang et al. (2022)
Agar (*Gelidium amansii Lamouroux*)	27.76%	Ji and Wang (2015); Zhang et al. (2022)

$$GE_y = GE_f + GE_{nf} + GE_{peay} + GE_{pepy} + GE_{feay} + GE_{fepy} + GE_b + GE_N + GE_{CH_4} \quad (13)$$

Finally, the annual EI can be calculated as Eq. (14), where P_y is the annual production.

$$EI_y = \frac{GE_y}{P_y} \quad (14)$$

2.7. Carbon fixation for shellfish and algae

Shellfish and algae are important carbon sinks in aquaculture. They can uptake a significant volume of coastal ocean carbon by removal of phytoplankton and particulate organic matter through filter-feeding, and seaweeds through photosynthesis (Gui et al., 2018). Although shellfish and algae emit few greenhouse gases, their greenhouse gases are so few that are ignored in this study. In this study, the weighing method was used for calculating carbon sink for shellfish and algae from 1991 to 2020. The formulas for shellfish are as follows:

$$C_B = C_{ST} + C_S \quad (15)$$

$$C_{ST} = P * R_{ST} * W_C \quad (16)$$

$$C_S = P * R_S * W_{CS} \quad (17)$$

$$C = 3.67 * C_B \quad (18)$$

C_B represents the fixed carbon (g), C_{ST} represents soft tissue carbon sequestration (g), C_S represents shell carbon sequestration (g). P: production of shellfish; R_{ST}: soft tissue dry mass ratio (%); W_C: soft tissue carbon content (%); R_S: shell tissue dry mass ratio (%); W_{CS}: shell tissue carbon content (%); C: the CO_2 uptake of shellfish (g). The shellfish species included in this study are shown in Table 1.

The formula for algae is as follows:

$$A = 3.67 * W_{CW} * \sum M \quad (19)$$

A represents the CO_2 uptake of algae (g), W_{CW}: the carbon content (%) of algae for different algal (dry weight), M: the dry weight of algae species (g) (Table 2), 3.67 is the conversion coefficient [44 (molecular weight of CO_2)/12 (molecular weight of C)]. The algae species included in this study are shown in Table 2.

3. Results

3.1. GHG emissions from China's marine aquaculture

From 1991 to 2020, the production of marine aquaculture (excluding shellfish and algae) increased continuously, reaching 3.94 Mt in 2020 (Fig. 2a). Along with the production, the total GHG emissions of the 10 provinces also increased continuously from 1.97 Mt (1991) to 24.96 Mt (2020) (Fig. 2a). It has increased by 1167.01% since 1991, with an average annual growth rate of 38.90%. Guangdong, as the largest province for GHG emissions, always had the highest annual GHG emissions (Fig. 2b), and had the largest proportion between 1991 and 2000. Fujian also contributed a high amount of GHG, and the proportion of Fujian is on the rise from 1991 to 2020. In contrast, Tianjin and Zhejiang provinces showed a downward trend in recent years, they contributed a smaller proportion to total GHG emissions (Fig. 2b). Among the five sectors, trash fish contributed the largest GHG emissions, besides, the commercial feed also had high GHG emissions, which showed that feed contributed the most GHG emissions. In contrast, CH_4 accounted for the smallest proportion of the total GHG emissions (Fig. 2c).

Among the 10 coastal provinces, Guangdong, Fujian, Shandong, and Liaoning provinces had the largest GHG emissions over the past 30 years (Fig. 3). Guangdong and Fujian provinces had the largest GHG emissions in 2020, which accounted for 34.91% (8.71 Mt) and 21.29% (5.31 Mt) of the total GHG emissions respectively. In contrast, Tianjin province had the smallest GHG emissions over the 30 years, which reached the largest emissions in 2012 (0.11 Mt), but still only accounted for 0.69% of the total emissions in that year (Fig. 3). Besides, Hebei province also had smaller emissions than other provinces, with the largest GHG emissions (0.65 Mt) in 2020 (Fig. 3).

Compared with other provinces, Guangdong always had the largest annual GHG emissions over the last 30 years. Otherwise, Tianjin province always had the smallest annual GHG emissions. Hebei had the highest proportion of GHG emissions in 1991, and its proportion got smaller and smaller. Although the proportion (1.21%) of GHG emissions for Guangxi was the smallest in 1991, the value had raised to 7.75% in 2020. According to Fig. 3, the 30-year proportional change in GHG emissions varies widely among the 10 provinces.

3.2. Emissions intensity of marine aquaculture

The EI of China's marine aquaculture for each year is shown in

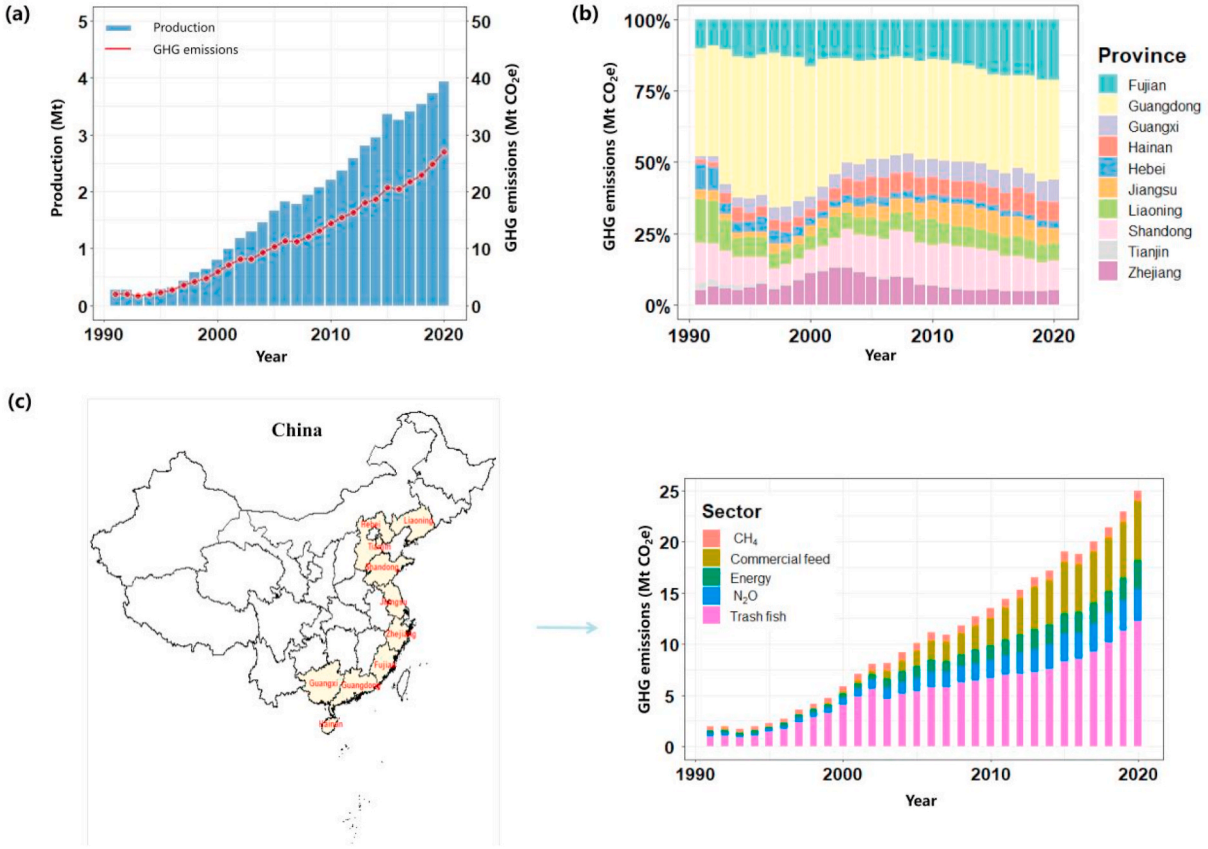


Fig. 2. Greenhouse gas (GHG) emissions from marine aquaculture over the past 30 years in China. (a) Trends in the volume of marine aquaculture production (excluding shellfish and algae) (Blue) and the amount of GHG emissions (Red) of marine aquaculture for the 10 provinces, 1991 to 2020; (b) the contribution of each province to the total amount of GHG emissions from marine aquaculture over the past 30 years, 1991 to 2020 (c) China map (provinces with white represent there are no collected data in this study) and the contribution of each sector (commercial feed, trash fish, N_2O, CH_4, and energy) to GHG emissions from 1991 to 2020.

Fig. 4a, which reached a maximum value of 9.49 in 1994, and a minimum value of 5.67 in 2015. The EI of marine aquaculture of the 10 provinces has shown a declining trend over the past 30 years. The trend of EI of the ten provinces increased from 1991 to 1994, then decreased from 1994 to 2015, and increased again during 2015–2020, but remaining at relatively lower values than in previous decades.

Besides, results showed that most provinces had EI < 10 (Fig. 4b). In contrast, a few provinces had very high EI values in specific years (e.g., the EI of Tianjin province where during 1993–1996 EI exceeded 15). Nevertheless, with the development of aquaculture, the values of provinces with higher EI gradually decreased, and the differences between provinces show a decreasing trend.

As shown in Fig. 4b, the 30-year EI varies greatly among provinces. Hebei and Liaoning had the largest values of EI. Tianjin province had the largest EI compared with other provinces in 2006, but its value decreased in 2020, which means that Tianjin province would emit less GHG in the same products compared with other provinces. Besides, the EI of Guangdong, Guangxi, Liaoning and Zhejiang provinces also decreased from 1991 to 2020, especially for Guangxi provinces, the value of which decreased from 11.28 to 4.26. However, the values of EI for Fujian and Jiangsu provinces showed fluctuations. Besides, Shandong and Hainan increased slightly in recent years (Fig. 4b).

3.3. Carbon sink by shellfish and algae

As shown in Fig. 5a, the production of shellfish and algae increased from 1991 to 2020 (1.27 Mt to 16.50 Mt). Meanwhile, CO_2-e was fixed by shellfish and algae from 0.44 Mt (1991) to 4.99 Mt (2020) (Fig. 5a).

When carbon sink was disregarded, the total GHG emissions from 1991 to 2020 increased from 1.97 Mt to 24.96 Mt, but GHG emissions could be reduced to 1.53 Mt and 19.97 Mt, respectively, when carbon sink is considered. Therefore, the "C sink effect" of shellfish and algae can be estimated as 4.99 Mt CO_2-e in 2020 (Fig. 5b). The EI dropped remarkably from 6.34 to 0.98 in 2020 due to C sink effect and increases in production (Fig. 5c).

As shown in Fig. 6, Shandong and Liaoning provinces were influenced by C sink effects rather than others. Moreover, GHG emissions from these provinces were negative in some years, reinforcing their role in C sinking. However, in Tianjin, Hainan and Guangdong provinces, without or with little production of shellfish and algae, GHG has not diminished greatly (Fig. 6). According to Fig. 6, EIs of the eight provinces (except Tianjin and Hainan provinces) had a marked decline. The EIs of Liaoning and Shandong provinces were almost null. Besides, GHG emissions in Hebei, Fujian, and Zhejiang provinces were also net zero due to C sink effect in some years (Fig. 6).

4. Discussion

4.1. Analysis of changes in GHG emissions

This study represents an important contribution to filling in the knowledge gap by providing, as far as we know, the first evidence of the links between large-scale aquaculture changes in production and GHG emissions of a country's marine aquaculture. Results showed that the trend of GHG emissions was close to the production over the past 30 years (Fig. 2a), and feed (including commercial feed and trash fish), as the main path of GHG emissions (crop energy use, crop LUC, fertilizer production, crop N_2O, rice CH_4 and others), contribute more GHG

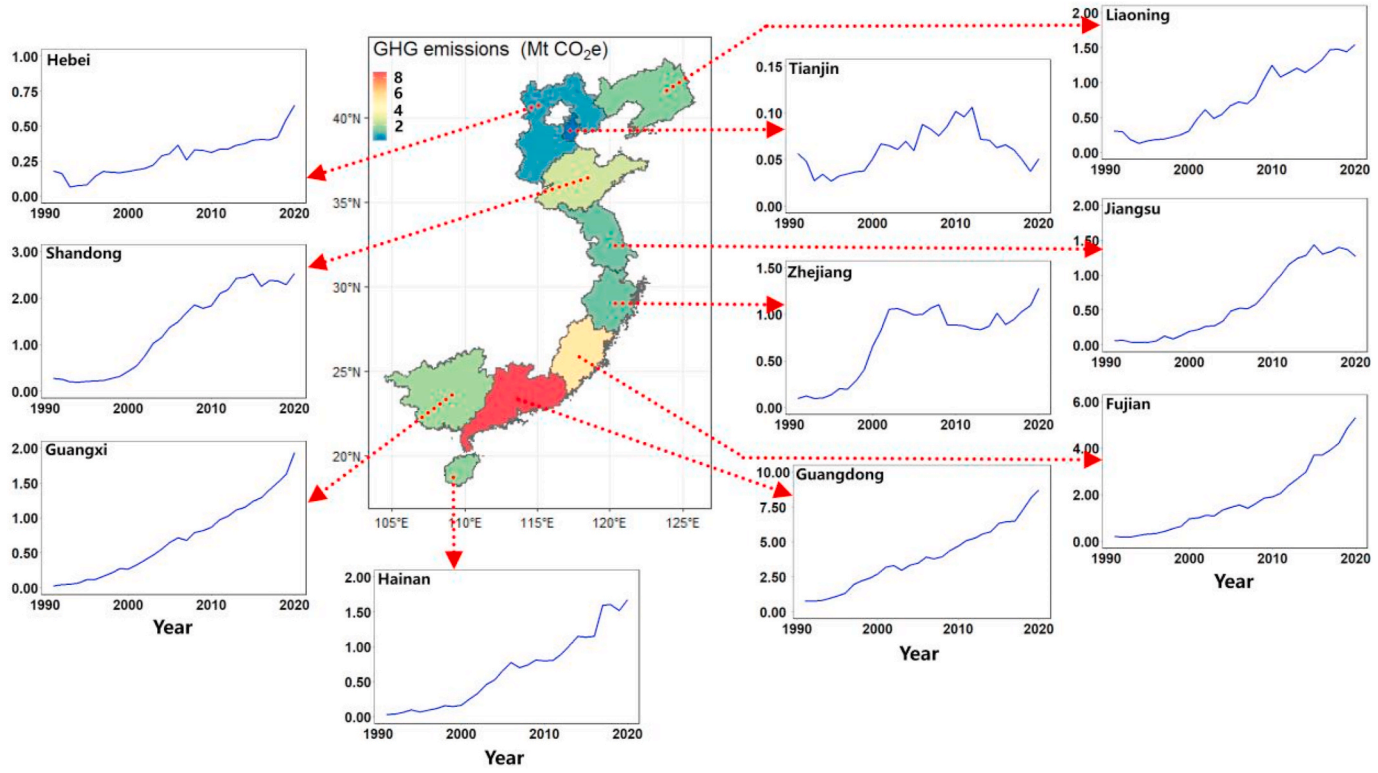


Fig. 3. The total amount of GHG emissions from marine aquaculture for each province in 2020 (map of 10 provinces in different colors) and time series of greenhouse gas emissions for each province over the past 30 years, from 1991 to 2020.

emissions (Fig. 2c) (Cao et al., 2011; MacLeod et al., 2019; Song et al., 2019). Consistent with our results, Adhikari et al. (2013) also found that feed accounts for about 90% of the total carbon equivalent of aquaculture inputs in Eastern India.

The dramatic changes in greenhouse gas emissions from marine aquaculture come from the rapid growth of marine aquaculture production. First, aquaculture is associated with household economy, income, and employment generation (Béné et al., 2016). From the 1990s, with the government subsidies to improve the living standards of rural people through increasing income, marine aquaculture development started to play a big role in increasing rural households' incomes in the coastal provinces due to the rich seawater source (Gui et al., 2018). Second, farming models, farming techniques and management are important factors in promoting the development of marine aquaculture (Gui et al., 2018; Jian, 2020; Zhao et al., 2021). Many scientific research projects have been conducted, which led to significant progress in aquaculture technology development, besides, breeding of species also make great progress in recent decades (Li et al., 2011). By establishing partnerships and fostering interactions between experts and stakeholders, the stakeholders can culture more scientifically according to the experts' suggestions (Gui et al., 2018). Third, public policies are also an important contributor to the development of marine aquaculture. For instance, the system of "Fish Seasons Closed" in 1995 and the 13th Five-Year Plan (FYP) echoes the call for "prioritizing ecosystem and promoting green development". They promote marine aquaculture while limiting the capture of wild fisheries (Zhao, 2021). Besides, a relatively comprehensive aquaculture extension system has been established under the leadership of the Ministry of Agriculture and Rural Affairs (Wang et al., 2020). Raul et al. (2020) also forecasted that GHG emissions will increase from 9.30×10^{10} g CO_2-e in 2009 to 3.83×10^{11} g CO_2-e in 2030 due to the increase in aquaculture production. Therefore, the rapid development of aquaculture provides people with more aquatic products, and more feed and energy use leads to more GHG emissions from multi-sectors.

Guangdong, Fujian, and Shandong provinces had the largest GHG emissions over the past 30 years, which could be related to the species that are cultured and the areas of marine aquaculture. Gephart et al. (2021) have shown that finfish and crustaceans can emit more GHG than other species, and Guangdong, Fujian, and Shandong provinces have proved that they are the major provinces for farming fish, shrimps, and crabs (Xu, 2020). The three species groups are the main species of consuming feed, so a large amount of feed is put into marine aquaculture in the three provinces.

4.2. *Multi factors contribute to the decline of emission intensity*

The results showed that the EI of marine aquaculture was declining (Fig. 4), which could be attributed to the continuous progress of aquaculture modes and the changes in feed (Li et al., 2011). The trend of EI representing the growth rate of GHG emissions is smaller than that of production. The rapid growth of production is not only from the increased areas but also comes from the changes in farming models. Marine aquaculture modes in China are currently classified into many types based on the type of practice (Gui et al., 2018). For instance, factory farming (such as flow-through and recirculating aquaculture systems) has greatly increased aquaculture production (Wang et al., 2020). Gui et al. (2018) have shown that factory farming can increases the output of unit farming area per hectare and leave more land to culture more production. In factory farming, the unit aquaculture production is relatively larger than in pond farming. From 2004 to 2020, the production of factory farming raised from 51,109 tons to 325,308 tons, and the density raised from 4.27 kg/cubic meter to 8.25 kg/cubic meter according to the China Fisheries Statistical Yearbook. The reduction in trash fish usage is another reason to reduce EI due to the higher EF of trash fish than that of commercial feed (MacLeod et al., 2020). Indeed, our results showed that the percentage of trash fish, which is used as non-commercial feed, continues to decline annually in China, paralleling the findings of Tang et al. (2016).

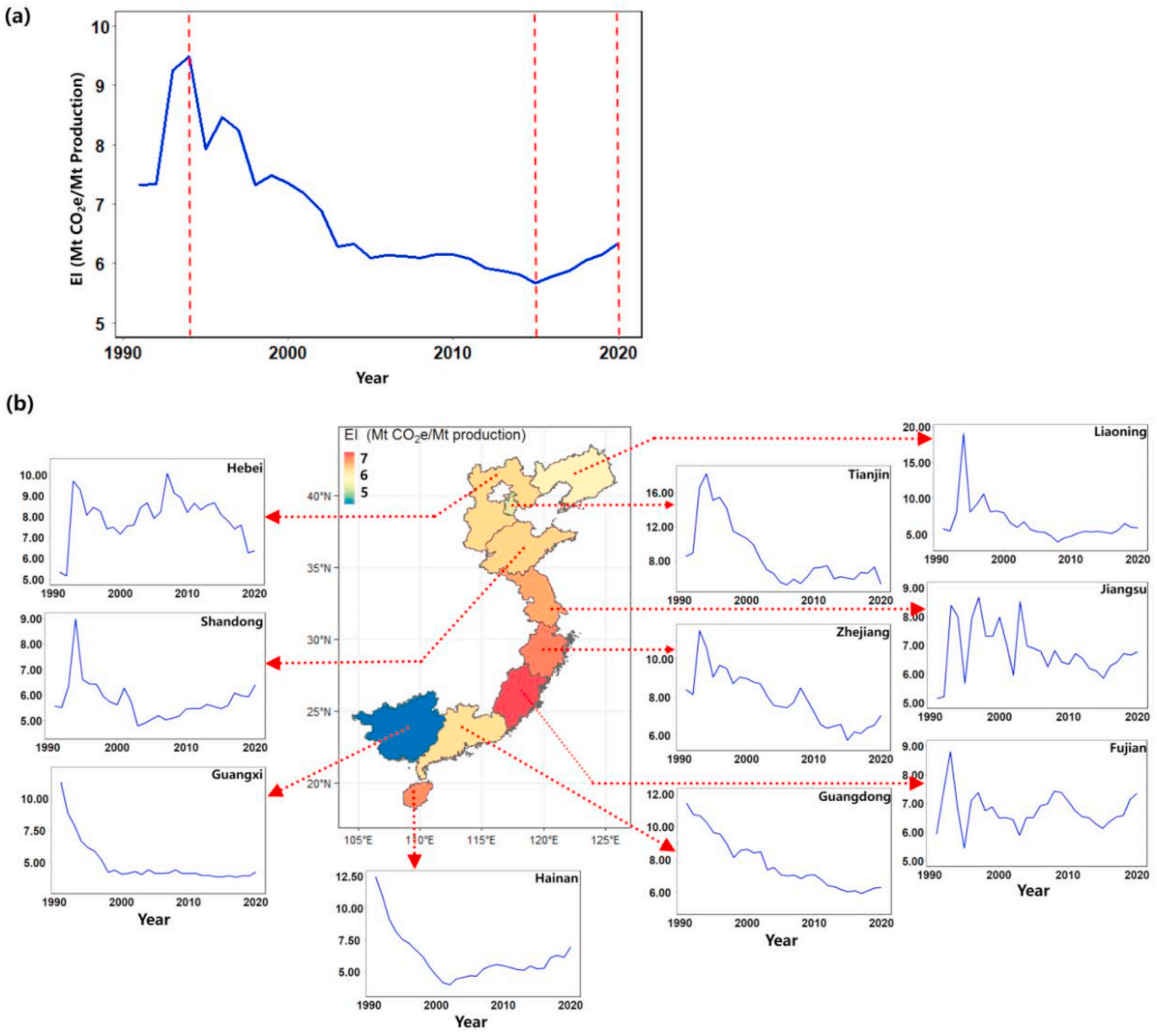


Fig. 4. Distribution of emissions intensity (EI) of greenhouse gas emissions from marine aquaculture over the past 30 years in China. (a) Trends of emissions intensity (EI, Mt CO_2e/Mt production) of marine aquaculture for the 10 provinces, 1991–2020; (b) emissions intensity (EI, Mt CO_2e/Mt production) of marine aquaculture for each province in 2020 (map of 10 provinces in different colors) and the time series of EI for each province over the past 30 years, 1991–2020.

Among the 10 provinces in this study, the EIs of Tianjin, Guangdong, Guangxi, Liaoning, and Zhejiang showed more promising trends than others, likely attributed to the above reasons. Shellfish and algae, as important carbon sinks, contribute to the balance between carbon emissions and carbon fixation (Sondaket al., 2017; Gui et al., 2018). As shown in Table 3, the production of shellfish and algae was small in Tianjin and Hainan, which only accounted for a small percentage of the provinces' production. In contrast, the two species groups accounted for more than 80% of production in Shandong, Liaoning, Zhejiang, and Fujian in 2020. Because shellfish and algae farming does not require arable land or crops, their feeds mainly come from the natural environment, and the food of shellfish comes directly from the water (Gentry et al., 2017; Ray et al., 2019; Alonso et al., 2020; Naylor et al., 2021). Besides, Gephart et al. (2021) proved that shellfish and algae have the least GHG emissions compared to other species. Therefore, the higher the proportion of shellfish production, the lower the GHG emissions. Moreover, shellfish produces larger quantities of food in similar areas than other species (Gentry et al., 2017; Froehlich et al., 2018; Ray et al., 2019).

4.3. Future directions and limitations

GHG emissions from the aquaculture sector are lower than terrestrial livestock (such as chickens, pigs and sheep) (Waite et al., 2014; Tacon and Metian, 2015). Meanwhile, this sector has great potential for further improvement. We found that China's marine aquaculture had lower emission intensity than the previous years, and contributed more aquatic products to people. As Bianchi et al. (2022) claimed that aquatic-sourced foods play an increasingly important role in the future diet, therefore, China can mitigate GHG emissions and increase the income of rural people by adjusting the proportion of species groups through export and domestic market and consumer diet structure. In addition, reducing energy use or changing the type of energy is also an important path to reduce GHG emissions, e.g., changing the current

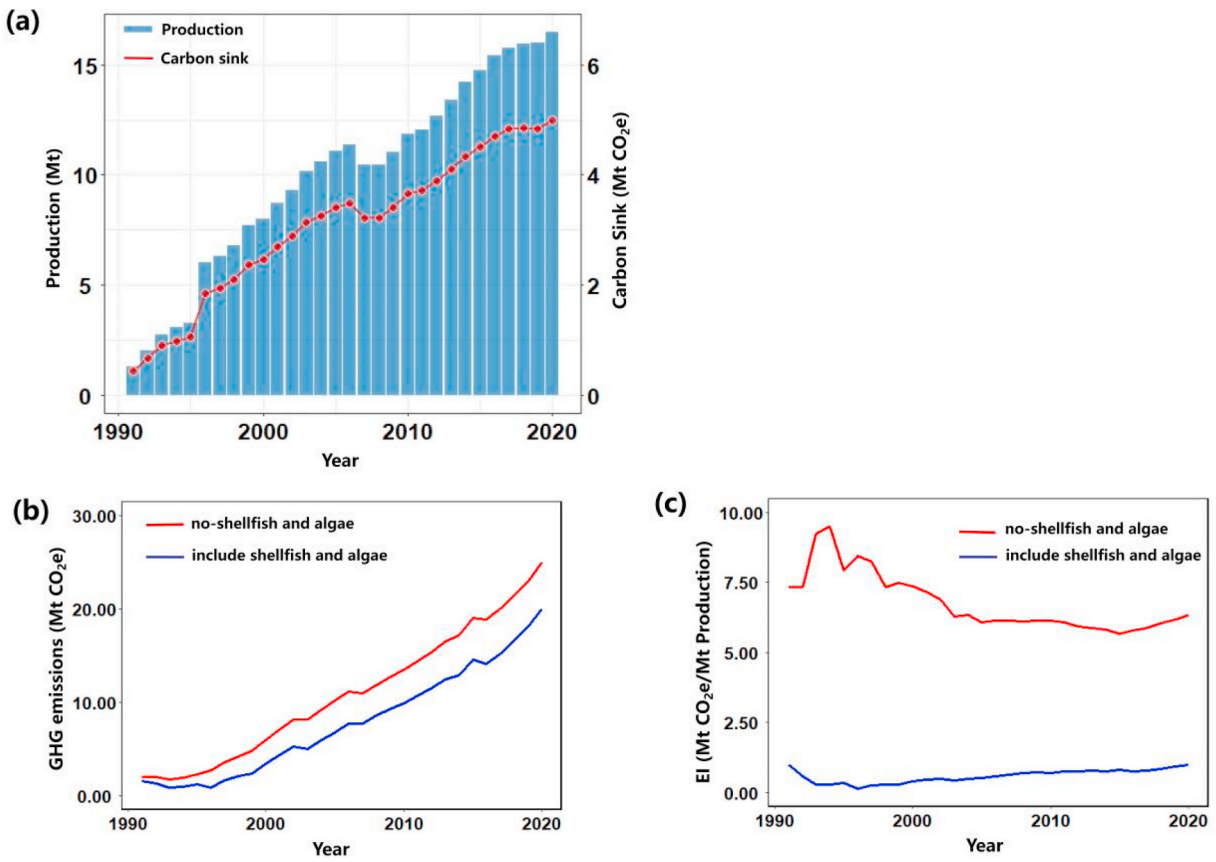


Fig. 5. Changes in greenhouse gas (GHG) emissions and emission intensity (EI, Mt CO_2-e/Mt production) when shellfish and algal were considered. (a) Trends in the volume of marine aquaculture production (shellfish and algae) (blue) in China, and the amount of Carbon sink (red) of shellfish and algae, 1991 to 2020; (b) greenhouse gas (GHG) emissions (Mt CO_2-e) for fish, shrimp, crabs, others (red) and fish, shrimp, crabs, others, shellfish and algae (blue); (c) emission intensity for fish, shrimp, crabs and others (red) and fish, shrimp, crabs, others, shellfish and algae (blue).

energy mix, using hydro, natural gas or nuclear power (Cao et al., 2011).

Finally, the government or society should establish effective recording measures so that changes in GHG emissions can be better analyzed. Due to the lack of information, we did not include the emissions from the post-farm supply chain, such as refrigerant leakage in cold chains and wastewater treatment. Therefore, the current estimates of GHG emissions might be underestimated. Nevertheless, the GHG emissions from the five sectors (i.e., commercial feed, trash fish, CH_4, N_2O and energy use) have been proven to be the major GHG emissions in the whole marine aquaculture process in many studies (Cao et al., 2011; Pelletier et al., 2009; Song et al., 2019). Therefore, this limitation will not affect our conclusions. In addition, the limitations of the parameters (such as the values of FCR are fixed for each species over the 30 years) did not affect the trend of the results (i.e., EI), because the values of FCR come from 2014, and GHG emissions might be underestimated before and overestimated after that year. But promoting accurate assessment of the sustainability of future farming technology, aqua-feed, and investments could be explored, we should collect more constant data to calculate GHG emissions of aquaculture across time. The spatiotemporal assessment will be helpful in terms of making targeted interventions and policy-making decisions to achieve the goals of sustainable food production.

5. Conclusions

This study is the first to assess systematically the changes in GHG emissions and EI over the past 30 years of marine aquaculture in China. With the rapid development of marine aquaculture, the output of marine aquatic products continues to increase therefore affecting GHG emissions. Even though the EI is decreasing, its variability between provinces gradually narrowed. Meanwhile, shellfish and algae as important carbon sinks, help to balance the carbon budgets of some provinces. Therefore, each province can achieve a balance by amending the ratio between cultured species, and adjusting the proportion of species production within coastal provinces to achieve carbon neutrality or even negative GHG emissions from China's marine aquaculture.

Author statement

Congjun Xu: Conceptualization; Formal analysis; Writing – original draft; Visualization; Guohuan Su: Conceptualization; Methodology; Formal analysis; Writing – review & editing; Supervision; Kangshun Zhao: Methodology; Formal analysis; Writing – review & editing; Huan Wang: Methodology; Formal analysis; Writing – review & editing; Xiaoqi Xu: Methodology; Formal analysis; Writing – review & editing; Ziqi Li: Methodology; Formal analysis; Data curation; Qiang Hu: Conceptualization; Methodology; Writing – review & editing; Jun Xu: Conceptualization; Methodology; Data curation; Writing – review & editing; Project administration; Supervision; Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

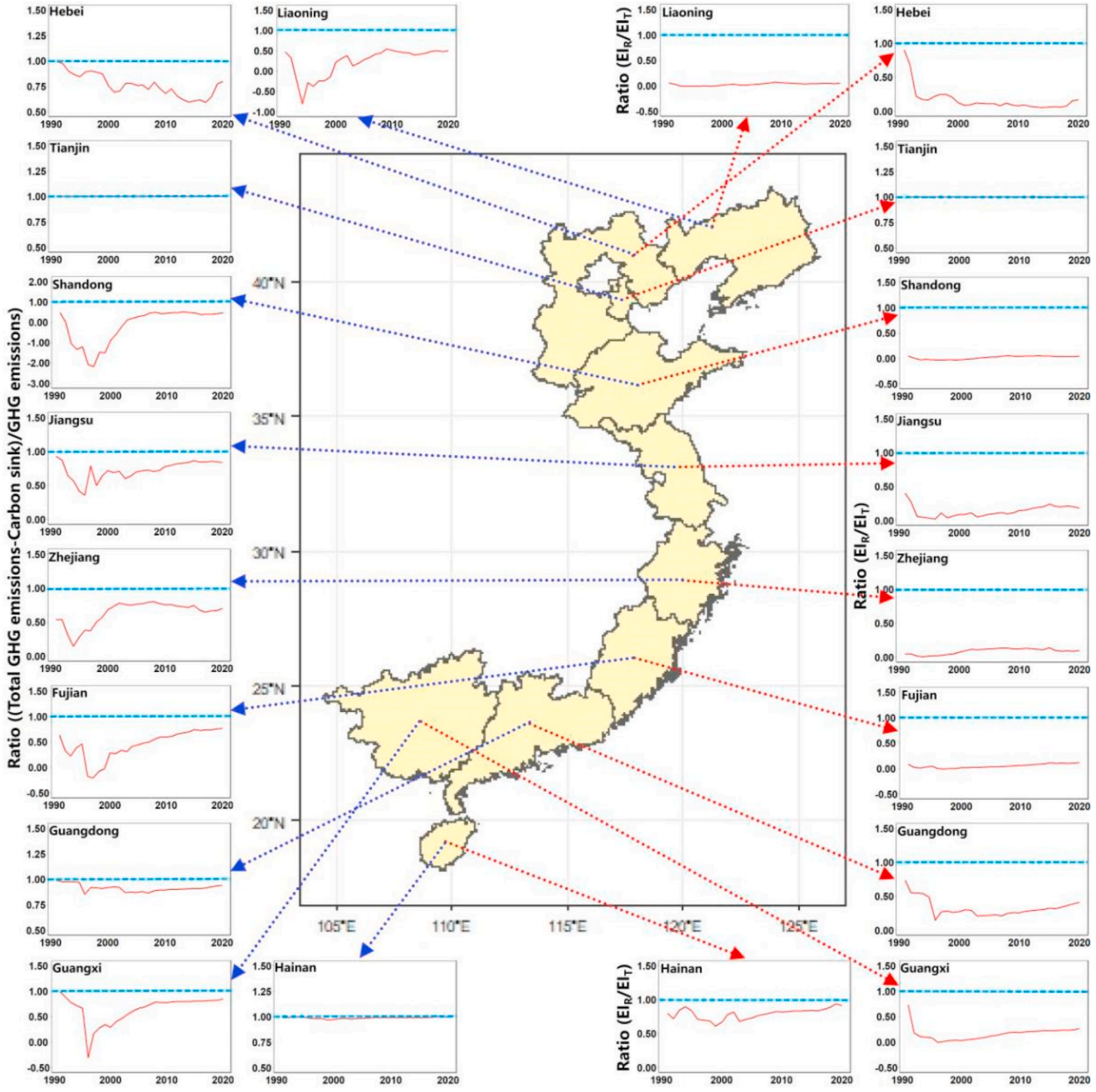


Fig. 6. Changes in greenhouse gas (GHG) emissions (blue arrow) and emission intensity (EI, Mt CO_2-e/Mt production) (red arrow) for each province when shellfish and algal were considered. The light blue line represented GHG emissions/GHG emissions = 1 and $EI_T/EI_T = 1$; EI_R: EI including shellfish and algae; EI_T: EI excluding shellfish and algae.

Table 3
Percentages of production of shellfish and algae for the six provinces in 2020.

Provinces	Tianjin	Hainan	Shandong	Liaoning	Zhejiang	Fujian
Percentage of shellfish and algae production	0.00%	8.65%	91.83%	90.93%	86.52%	85.82%
Percentage of other production	100%	91.35%	8.17%	9.07%	13.48%	14.18%
Sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Note: shellfish and algae include the main cultured species. Other production include fish, crabs, shrimps, sea cucumber, jellyfish et al.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2022.117025.

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